

When Should the Method Win—and Lose?

Directional hypotheses and algebraic cases; no fitted method ranking.

The objective is future team quality at complete cost, with reliable attribution and timely updates. Reasonable expectations follow a mechanism and a matched contrast. We do not assign a loss merely to make a results table look credible. This guide supplements v2's exact conditional distribution model with conditions, strong controls, adverse cases and falsifiers for the three body tables.

Independent review: the full forecast gate has not passed

Review dimension	Verdict	Evidence or missing requirement
Conditional arithmetic	Pass in scope	v2's same-outcome means, covariance and finite-count errors; assumed inputs.
Benchmark fit/authority	Partial / open	ArtifactRole fits the handoff question; distinct qualified roots and native later-use bridge remain open. PeerSelect is a separate adaptation track.
Baseline fairness	Open	Same information, task-dependent matching, eligibility, exploration and complete cost must be qualified live.
Empirical means/errors	Not identified	No qualified method-level future stream population or fitted uncertainty.
Advantage/weakness logic	Repaired in v3	Two-sided conditions below; superiority still requires actual outcomes.

Existing evidence: snapshots, not a forecast sample

The C1 snapshot cited by v2 has eight complete real API calls on one PIPE3 root, no executed task after its updates, and a positive judgment despite failing producer/terminal checks. Its 0.520/0.297 ms timings are single core updates. The separate October 8 diagnosis made two real judgment calls, stopped on a producer false acceptance, and left two cells unstarted. A public execution trace exposes the datetime space/T mismatch; model use of that trace is untested. Neither snapshot estimates judgment error rates or future method gains.

Table 1 baseline roles: a method must beat the right alternative

Body-table row / control	What the contrast must rule out
Uniform / no update	Exploration or base prior alone; cold start can tie with more overhead.
Raw acceptance / terminal only	Judgment content beyond raw verdicts or independent outcomes; neither is privileged truth.
Contextual trust linear	Ordinary contextual learning with the same public information; no presumed ranking.
Task-conditioned ridge*	Ordinary task-peer preference reversal; additive identity features alone are insufficient.
Pooled controller	Benefit of local personalized state versus pooling; sparse histories may favor pooling.
Meta-Team L2-public	Closest published public-state alternative; adapter remains unqualified.
RARE / matched composition*	Added mechanism beyond identical representation, gates and ordinary update/retrieval components.

*Diagnostic comparators supplement the eight Table 1 rows; this guide does not silently change the active paper matrix. Ridge has offline algebraic controls, not live parity. Published native scores constrain metric definitions and plausible scale, not our ranking: TeamBench partial .7544 is not binary pass; Meta-Team's 13.1 pp system gain is not its 3.2 pp L2 increment. The versioned reference archive preserves conflicting results.

Table 1: Information and Future Allocation**Directional hypotheses and algebraic cases; no fitted method ranking.**

Quality means an independently scored *later* task. $U = Y - \lambda C$ uses all declared costs. A changed assignment is a process diagnostic, not a quality goal; frequent changes can be harmful. Every direction below is conditional.

Condition	Matched contrast	Plausible gain and possible loss	Falsifying observation
Varied, informative judgments	Full vs count-only, J-masked, Qp-only and terminal-only	Lower held-out prediction loss and better later allocation <i>if</i> J adds information. Extra judgment/retrieval cost may erase utility gain.	Masking J preserves the gain, or predictions improve but later quality does not.
Public execution facts explain the defect	Trace+J vs trace-only; J-masked	Added explanation may help transfer, but trace-only may tie or win at lower cost. No assumed J increment.	All improvement is reproduced by the same execution trace without J.
Stable task-peer crossover	Full vs task-conditioned ridge and matched composition	Context-specific matching may beat uniform/global trust. Advantage over the strong task-conditioned control is unknown.	Only weak additive controls lose, or ridge explains the whole gain.
Uninformative or constant J	Full vs no-update, count-only and strong prior	No J-specific gain is justified. If choices/outcomes are identical and extra cost is positive, full utility is lower.	Claimed J benefit persists after masking/permuting with valid support.
Incorrect, stale or mismatched evidence	Full vs static prior and same-info strong control	Learned evidence can reduce quality; costs and correction lag can worsen utility. A robust control may do better.	Benefit depends on wrong peer/version matching, leaked truth or future information.
Sparse comparable history / new root	Full vs pooled, prior and ridge; 0/1/K support	Personalized history may help once supported; pooling may win with few labels. Cold-start overhead and negative transfer are realistic risks.	Gain disappears under root/time holdout or only follows repeated task templates.

Fairness is part of the mechanism test

Match the task, candidate@version/menu, public execution facts, feature/encoder coverage, read cutoff, selected-only feedback, delay, eligibility/UNKNOWN, propensity and exploration, capacity and complete service budget. Separate two interventions: removing a responsibility gate tests that gate; changing the updater while keeping the gate tests the updater. Do not give only full RARE the execution trace or stronger task features.

Prediction comparisons cannot use different selected populations

Compare probability forecasts on a common, sealed evaluation population with legally acquired real labels and specified sampling weights/propensities. Keep that evaluation separate from training feedback. Otherwise a policy may change its own Brier by routing to easier cases. Choice propensity is not predicted correctness. In v2, $p = q + e$ is a latent-oracle sensitivity case, not a deployable predictor; S2/S3 Brier equality follows the symmetric .7/.3 assumption and is not a general routing property.

Tables 2–3: Adaptation, Safety and Mechanisms

Directional hypotheses and algebraic cases; no fitted method ranking.

Table 2: fast updates do not guarantee low end-to-end latency

Condition	Comparator	Two-sided expectation / required measurement
Stable peers, little novelty	No history / static prior / ordinary updater	Full may tie quality and lose on service/state cost. Measure whole-service p50/p95, state and complete cost; core time alone is insufficient.
Gradual drift, timely legal labels	Frozen prior / same-info RLS, SGD or refit	Online feedback may recover earlier than frozen state. A strong updater may match or beat full; retention can slow adaptation. Measure old/new holdouts, recovery and utility.
Abrupt change / peer replacement	Static prior / reset / version-aware control	Stale profiles may cause negative transfer; version isolation may avoid it but restart learning. Report recovery, old-task loss and feedback coverage.
Long feedback delay / few eligible outcomes	History-only / matched delayed updater	Public observations may help early decisions if predictive; persistent training still waits for legal later credit. Sparse credits can make adaptation slower.
High arrival rate / expensive evidence	Local uniform / matched service stack	Selection quality can rise while backlog and p95 worsen. Measure arrivals, full service distribution, queueing, drops/timeouts and net payoff.

Graph-IPD is a native multi-round cooperation reference, not an ArtifactRole score or an online service result. Its published cooperation bounds cannot populate payoff, update p95, forgetting or recovery cells. Timing, drift and retention methods are not finally locked; these are tests, not feature claims.

Table 3: attribution safety and organic interaction

Contrast	What can improve / what can worsen	Disconfirming outcome
Gate vs no gate	Independent owner truth can reduce false producer credit. Abstention costs coverage, usable labels and update speed. Legal positive/negative J remain observations; gate is not accept-only.	Same safety but less coverage and worse future utility; factually wrong J still treated as reward.
Public evidence $\times$ delayed credit (four cells)	Early public information plus later correction may work together. Neither interaction nor dominance of the full cell is automatic.	No joint increment, or full loses to one component or matched composition.
Local vs nonlocal	Locality can lower communication/cost but miss a better distant peer. Compare complete costs and quality; no universally best neighborhood.	Cost saving comes entirely from a smaller information/compute budget.
Correction/replay vs ordinary state	Exact identity/idempotence can prevent duplicate credit; retraction/retraction has state and service costs. Engineering safety is not an efficacy estimate.	Legal credit counted twice, correction changes the wrong version, or gains vanish with matched replay.

Four-cell utility interaction is $I_U = U_{11} - U_{10} - U_{01} + U_{00}$, with $\text{Var}(\hat{I}_U) = c^\top \Sigma c$ for $c = (1, -1, -1, 1)$ in that cell order. Report the qualified stream covariance; do not assign four convenient SDs. Zero interaction does not by itself negate the value of components, but fails a claimed synergy. All adaptive arms keep their own actual histories.

Quantitative Boundaries and Distribution Acceptance

Directional hypotheses and algebraic cases; no fitted method ranking.

A quality gain can lose after complete cost

$$\Delta U = \Delta \mathbb{E}[Y] - \lambda \Delta \mathbb{E}[C], \quad \Delta U > 0 \iff \Delta Q > \lambda \Delta C.$$

Case: $\lambda = .1$ below is the v2 *illustrative* weight, not a frozen scientific acceptance margin. Cost is normalized complete cost. These are algebraic boundary inputs, not fitted means, predictions or method ranks.

Case	Explicit assumption	ΔQ	ΔC	ΔU
B0	No outcome gain, extra work	+0.000	+0.100	-0.010
B1	Gain exceeds overhead	+0.040	+0.100	+0.030
B2	Exact break-even	+0.020	+0.200	-0.000
B3	Small gain, costly evidence	+0.020	+0.300	-0.010
B4	Misleading or stale evidence	-0.020	+0.050	-0.025
B5	Better but expensive peer	+0.060	+0.800	-0.020
B6	Gain with fewer repairs	+0.020	-0.100	+0.030

No case is assigned to RARE or a baseline. Observed methods may cross these boundaries; that is exactly what the complete-cost experiment must determine.

Why v2's repair-only cost law is optimistic for routing

Within v2 only, $\Delta U = (1 + \lambda\gamma/c_0)d\Delta\kappa - (\lambda/c_0)\Delta h$. With $d = .2$, $\gamma = 1$, $c_0 = 1.5$ and $\lambda = .1$, positive utility requires $\Delta\kappa > 0.3125\Delta h$. Case: the assumed S3–S2 routing difference .25 reverses only when extra raw overhead exceeds 0.800. This weak penalty is not empirically justified. Real selection, judging, retrieval, publication, repair, failures and resource use must enter the cost ledger; stronger peers may be more expensive even when successful. Refit the model rather than forcing reality into its negative quality–cost correlation.

UNKNOWN can reverse an apparent quality ranking

For all assignments, let m be the fraction with unknown binary outcome and q_{obs} the success rate among known outcomes. Without a missingness model,

$$Q \in [(1 - m)q_{\text{obs}}, (1 - m)q_{\text{obs}} + m].$$

Case: hypothetical candidate .65 with $m = .15$ gives $[0.5525, 0.7025]$; control .60 with $m = .02$ gives $[0.5880, 0.6080]$. These overlap, so the higher observed rate does not establish a robust advantage. The bounds are logical limits, not confidence intervals; sampling uncertainty still applies. Unscorable resource failures remain UNKNOWN, with their attempted costs and denominators retained.

Means, errors and distributions must be validated together

1. Seal qualified roots, independent live histories, same-info comparators, primary quality/cost metrics, budgets and stopping rules before a pilot. Fit on development only; validate on untouched future streams/root holdouts.
2. Model the joint observable outcomes and complete costs. Utility variance is $\sigma_U^2 = \sigma_Q^2 + \lambda^2\sigma_C^2 - 2\lambda\text{Cov}(Q, C)$. High quality–utility correlation is not independent corroboration. No guessed per-cell SD or 95% interval is supplied here.
3. Report residual bias, root heterogeneity, serial correlation, covariance, 80/95% prediction coverage *and* width with uncertainty, alongside UNKNOWN by arm/root/J/owner. Prediction intervals differ from confidence intervals of a treatment effect; small root counts cannot justify a normal 1.96 band automatically.
4. Test the favorable and adverse conditions above. A strong control explaining the whole gain, or complete cost removing it, triggers diagnosis and method iteration. It does not authorize rewriting Goal or discarding failure logs.